

SQL AI Optimizer ■

Project Overview

An AI-powered SQL Query Optimization platform that analyzes SQL queries, detects performance bottlenecks, recommends indexes, and generates AI-based optimization insights using Gemini API.

Features

- SQL Query Analysis
- Execution Plan Visualization
- AI-Powered Optimization Suggestions
- Performance Bottleneck Detection
- Index Recommendations
- User Authentication (MongoDB)
- PostgreSQL Integration
- Modern React + Tailwind Frontend

Tech Stack

Frontend: React.js, Vite, Tailwind CSS, Axios, React Router DOM

Backend: FastAPI, Gemini API, PostgreSQL, MongoDB, PyMongo, Psycopg2

Environment Variables

```
GEMINI_API_KEY=your_gemini_api_key  
DATABASE_URL=your_postgresql_url  
MONGO_URL=your_mongodb_url
```

Run Frontend

```
cd frontend
```

```
npm install
```

```
npm run dev
```

Run Backend

```
cd backend
```

```
pip install -r requirements.txt
```

```
uvicorn app.main:app --reload
```

Deployment Links

Frontend: <https://your-frontend-link.onrender.com>

Backend: <https://your-backend-link.onrender.com>

API Endpoint

POST /optimize

Sample Request Body

```
{ "query": "SELECT * FROM orders WHERE customer_id = 5" }
```

Future Enhancements

- Query History
- Dashboard Analytics
- Query Cost Graphs
- AI Chat Assistant
- Multi-Database Support
- JWT Authentication

Author

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